

Matthew Tran

Linux Command Line Practice (BASE)

This allowed me to set up a base for my future activities using Linux. The purpose of this lab allowed me to practice essential terminal skills by navigating, creating, and managing files and directories using basic Linux commands.

Tools Used

- Linux terminal
- Bash shell

Skills Learned

- Customizing the terminal prompt using export PS1
- Navigating the filesystem with cd, pwd, and ls
- Using both absolute and relative paths to move between directories
- Listing and filtering directory contents with ls, ls -l, and ls -d
- Viewing file contents with tail, file, and tail -n+
- Creating, copying, moving, renaming, and deleting files and directories using mkdir, cp, mv, touch, and rm
- Performing file and directory management with superuser privileges using sudo

Task 1) Customize Terminal Prompt

I opened a terminal and used **export PS1="Matthew Tran@vm"** to personalize my command prompt. This makes it easier to identify my terminal session in a multi-user environment.

```
[09/09/24] seed@VM:~$ export PS1='matthew_tran@vm:\w$ '
matthew_tran@vm:~$
```

Task 2) Identify Current Location and Directory Contents

I used **pwd** to confirm my current location in the filesystem (/home/student), establishing my starting point. Then, I ran **ls** to list the contents of the directory, which helped me understand what files and folders were available before continuing.

```
matthew_tran@vm:~$ pwd
/home/seed
matthew_tran@vm:~$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos
matthew_tran@vm:~$
```

Task 3) Navigate Using an Absolute Path.

I used **cd /etc** to navigate directly to the system configuration directory. This demonstrates how absolute paths allow movement from anywhere in the filesystem to a specific location.

```
matthew_tran@vm:~$ cd /etc
matthew_tran@vm:/etc$
```

Task 4) Navigate Using a Relative Path

From /etc, I ran **cd skel** to enter the skeleton directory. This shows how relative paths enable navigation from the current directory without retyping the full path.

```
matthew_tran@vm:/etc$ cd skel
matthew_tran@vm:../skel$
```

Task 5) Move to Parent Directory

I used **cd ..** to move back to the parent directory (/etc). This is needed for efficient directory traversal without memorizing full paths.

```
matthew_tran@vm:../skel$ cd ..
matthew_tran@vm:/etc$
```

Task 6) List Directory Contents

I ran **ls** to view all files and subdirectories in **/etc**. This command is fundamental for exploring directory contents and understanding system structure.

Task 7) See More Details About Files

I used **ls -l** to see extra info like who owns each file, when it was made, and how big it is.

```

total 1164
drwxr-xr-x 3 root root 4096 Jul 31 2020 acpi
-rw-r--r-- 1 root root 3028 Jul 31 2020 adduser.conf
drwxr-xr-x 3 root root 4096 Jul 31 2020 alsa
drwxr-xr-x 2 root root 4096 Nov 24 2020 alternatives
-rw-r--r-- 1 root root 401 Jul 16 2019 anacrontab
-rw-r--r-- 1 root root 433 Oct 1 2017 apg.conf
drwxr-xr-x 5 root root 4096 Jul 31 2020 apm
drwxr-xr-x 3 root root 4096 Jul 31 2020 apparmor
drwxr-xr-x 7 root root 4096 Nov 24 2020 apparmor.d
drwxr-xr-x 4 root root 4096 Nov 24 2020 appport
-rw-r--r-- 1 root root 769 Jan 18 2020 appstream.conf
drwxr-xr-x 7 root root 4096 Nov 24 2020 apt
drwxr-xr-x 3 root root 4096 Jul 31 2020 avahi
-rw-r--r-- 1 root root 2319 Feb 25 2020 bash.bashrc
-rw-r--r-- 1 root root 45 Jan 25 2020 bash_completion
drwxr-xr-x 2 root root 4096 Nov 24 2020 bash_completion.d
-rw-r--r-- 1 root root 367 Apr 14 2020 bindresvport.blacklist
drwxr-xr-x 2 root root 4096 Apr 22 2020 binfmt.d
drwxr-xr-x 2 root root 4096 Jul 31 2020 bluetooth
-rw-r----- 1 root root 33 Jul 31 2020 brlapi.key
drwxr-xr-x 7 root root 4096 Jul 31 2020 brltty
-rw-r--r-- 1 root root 26916 Mar 3 2020 brltty.conf
drwxr-xr-x 3 root root 4096 Jul 31 2020 ca-certificates
-rw-r--r-- 1 root root 6505 Nov 24 2020 ca-certificates.conf
-rw-r--r-- 1 root root 5714 Jul 31 2020 ca-certificates.conf.d
pkg-old
drwxr-xr-x 2 root root 4096 Jul 31 2020 calendar
drwxr-s--- 2 root dip 4096 Nov 24 2020 chatscripts
drwxr-xr-x 2 root root 4096 Nov 24 2020 console-setup
drwxr-xr-x 2 root root 4096 Jul 31 2020 cracklib

```

Task 8) Only Show Files Starting with 's'

I typed `ls -d s*` to only list items in `/etc` that start with the letter "s".

```

matthew_tran@vm:/etc$ ls -d s*
sane.d          services        skel            ssl             sudoers
security        sgml            smi.conf        subgid          sudoers.d
selinux         shadow          snmp            subgid-         sysctl.conf
sensors3.conf  shadow-        speech-dispatcher subuid          sysctl.d
sensors.d       shells         ssh             subuid-         systemd

```

Task 9) Find Out What Kind of File `/etc/group` Is

I used `file /etc/group` to check what type of file it is.

```

matthew_tran@vm:/etc$ file /etc/group
/etc/group: ASCII text

```

Task 10) Look at the Last 5 Lines of a File

I typed `tail -n 5 /etc/group` to see only the last 5 lines of the `/etc/group` file.

```
matthew_tran@vm:/etc$ tail -n 5 /etc/group
systemd-coredump:x:999:
vboxsf:x:133:
telnetd:x:134:
ftp:x:135:
docker:x:136:seed
```

Task 11) View a File Starting at Line 20

I used `tail -n+20 /etc/passwd` to look at the `/etc/passwd` file starting from line 20 to the end.

```
matthew_tran@vm:/etc$ tail -n+20 /etc/passwd
systemd-resolve:x:101:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
systemd-timesync:x:102:104:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin
messagebus:x:103:106::/nonexistent:/usr/sbin/nologin
syslog:x:104:110:/home/syslog:/usr/sbin/nologin
apt:x:105:65534::/nonexistent:/usr/sbin/nologin
tss:x:106:111:TPM software stack,,,:/var/lib/tpm:/bin/false
uidd:x:107:114:/run/uidd:/usr/sbin/nologin
tcpdump:x:108:115::/nonexistent:/usr/sbin/nologin
avahi-autoipd:x:109:116:Avahi autoip daemon,,,:/var/lib/avahi-autoipd:/usr/sbin/nologin
usbmux:x:110:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
rtkit:x:111:117:RealtimeKit,,,:/proc:/usr/sbin/nologin
dnsmasq:x:112:65534:dnsmasq,,,:/var/lib/misc:/usr/sbin/nologin
cups-pk-helper:x:113:120:user for cups-pk-helper service,,,:/home/cups-pk-helper:/usr/sbin/nologin
speech-dispatcher:x:114:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
avahi:x:115:121:Avahi mDNS daemon,,,:/var/run/avahi-daemon:/usr/sbin/nologin
kernoops:x:116:65534:Kernel Oops Tracking Daemon,,,:/usr/sbin/nologin
saned:x:117:123:/var/lib/saned:/usr/sbin/nologin
nm-openvpn:x:118:124:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
hplip:x:119:7:HPLIP system user,,,:/run/hplip:/bin/false
whoopsie:x:120:125::/nonexistent:/bin/false
colord:x:121:126:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
geoclue:x:122:127:/var/lib/geoclue:/usr/sbin/nologin
```

Task 12) Go Back to Your Home Folder

I typed `cd ~` to go straight back to my home folder

```
matthew_tran@vm:/etc$ cd
matthew_tran@vm:~$ █
```

Task 13) Make a New Folder and Check It

I made a folder called data with **mkdir data**, then used ls to see that it was created.

```
matthew_tran@vm:~$ ls
data      'dmesg tail'  Downloads  Pictures  Templates
Desktop   Documents     Music      Public   Videos
```

Task 14) Copy a System File into Your Folder

I copied **/etc/passwd** into my data folder using **cp /etc/passwd data/**.

```
matthew_tran@vm:~$ cp /etc/passwd data
matthew_tran@vm:~$ ls data passwd
ls: cannot access 'passwd': No such file or directory
data:
passwd
```

Task 15) Copy a Whole System Folder

I used **sudo cp -r /etc/ppp .** to copy the entire **/etc/ppp** folder into my current location. I needed **sudo** because it's a protected system folder.

```
Matthew Tran@vmsudo cp -r /etc/ppp /home/student
Matthew Tran@vm ls
data      Documents  Pictures  Templates
Desktop   Downloads  ppp       Videos
'dmesg tail' Music      Public
```

Task 16) Rename a Folder

In the screenshot above, I used “:wq” to save the file and exit vi. Next, I will move the test file into the data directory using mv.

Task 18) Remove a File

I deleted the passwd file from the data folder using `rm data/passwd`.

```
Matthew Tran@vmvi test
Matthew Tran@vm mv test data
Matthew Tran@vmls data
passwd test
```

Task 19) Remove a Folder and Everything Inside

I deleted the peers folder and all its contents with **rm -r peers**.

```
Matthew Tran@vm sudo rm peer -r
Matthew Tran@vmls
data      'dmesg tail'    Downloads  Pictures  Templates
Desktop   Documents  Music      Public    Videos
```

Every command I ran had a clear job, like moving, checking, or cleaning up.